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RESEARCH ARTICLE

Bidirectional Haptic Feedback for Prostheses via Flexible Sensing and Transcutaneous Electrical Stimulation

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ABSTRACT

Upper-limb amputation disrupts natural somatosensory pathways and impairs motor control, increasing reliance on visual feedback during prosthesis operation and reducing intuitiveness and embodiment. Although haptic feedback can improve controllability, most commercial myoelectric prostheses still lack effective closed-loop feedback. Here, we present a flexible Bidirectional Haptic Feedback System (BHFS) integrating triboelectric multidimensional tactile sensing with multichannel transcutaneous electrical stimulation (TES). The customized Triboelectric Flexible Tactile Sensor (TFT-Sensor), featuring a strontium titanate ($SrTiO_3$)-modified triboelectric layer with a pyramid microstructure, achieves a 75% increase in open-circuit voltage compared to the baseline, enabling zero-power sensing of pressure, shear force, and surface texture. A flexible electrode armband driven by a multichannel electrical stimulator delivers spatiotemporally encoded electrical patterns through closed-loop mapping algorithms, providing intuitive multidimensional somatosensory feedback. In simulated prosthetic-hand experiments under audiovisual deprivation, subjects successfully adjusted grip force in real time, maintained stable grasping, and distinguished different surface textures. By establishing a closed-loop "sensing-mapping-stimulation" framework, the proposed system offers a promising strategy for restoring naturalistic somatosensation in prosthetic devices and advancing rehabilitation engineering and human–machine interfaces.

1 | Introduction

Upper-limb amputation results in not only permanent physical loss but also disruption of somatosensory pathways and motor control, compelling users to rely heavily on visual cues and prior experience during grasping, object manipulation, and fine motor tasks [1–6]. This long-term reliance on vision and the external shift of attention significantly increases cognitive load, diminishes operational intuitiveness and embodiment, and ultimately compromises independence in daily living [7, 8]. Although recent advances in prosthetic mechanics and actuation have partially restored motor function [9–11], the absence of tactile

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input remains a critical barrier to natural and efficient human-machine collaboration [12–16]. Without access to force perception and error-correction pathways comparable to those of a biological hand [17], users are prone to excessive grip force, object slippage, and frequent failures, undermining operational confidence and long-term adoption [18]. Moreover, most commercial myoelectric prostheses still operate primarily under open-loop control, falling short of achieving precise and natural interactive performance [9, 19].

At the research level, studies have demonstrated that incorporating haptic feedback into prosthetic systems can elicit corresponding neural responses and effectively improve controllability and grasp stability [10, 11, 20–23]. For instance, Jorgovanovic et al. systematically reviewed sensory substitution strategies and their benefits for prosthetic performance and user experience [24]. Luo et al. developed a closed-loop system based on single-pressure sensing and vibrotactile feedback to assist in basic grasping tasks [25]. Gu et al. created a soft neuroprosthetic hand integrated with capacitive tactile sensing and electrical stimulation feedback, leveraging intrinsic compliance to enable safe interaction with fragile objects [26]. Shen et al. developed an ionogel-based supercapacitive tactile skin with high sensitivity and a broad pressure detection range, providing reliable touch sensing for prostheses [27]. Xu et al. reported a self-powered textile-based electrotactile glove that harnesses daily hand-motion-induced triboelectric effects and gas breakdown discharge to deliver localized electrical stimulation, offering a new paradigm for thin, self-sustaining, and portable haptic feedback [28]. These approaches, by establishing artificial sensory channels on the residual limb skin, have preliminarily validated the effectiveness of closed-loop feedback.

However, current tactile feedback prototypes for prosthetic hands still face three common limitations (see Tables S1–S4 for comparison): First, sensing dimensionality is restricted—most systems focus on normal pressure, failing to capture tangential forces, slip, or surface texture, which are critical for safe and dexterous manipulation [29–42]. Second, feedback mapping strategies are underdeveloped—the relationship between tactile features and stimulation parameters remains simplistic, resulting in limited dimensionality and naturalness of the feedback [43–49]. Third, system-level integration and generalizability remain limited, as bulky components, complex wiring, high power consumption, and limited long-term wearability hinder real-world adoption [19, 50–52]. Therefore, there is a pressing need for a system-level closed-loop solution that concurrently fulfills the requirements of multidimensional sensing, natural feedback mapping, and high integrability.

To address these challenges, this study presents a flexible Bidirectional Haptic Feedback System (BHFS) that integrates multidimensional triboelectric tactile sensing with transcutaneous electrical stimulation (TES) for amputees. The system establishes a closed-loop "sensing-mapping-stimulation" cycle, aiming to achieve systematic restoration from fine tactile capture to natural somatosensory feedback. We developed a Triboelectric Flexible Tactile Sensor (TFT-Sensor) that exhibits a 75% increase in open-circuit voltage over the baseline, achieved through functional material modification and microstructure optimization. Multiple sensor units are integrated into a flexible Tactile Perception Glove (TP-Glove) to capture multidimensional tactile information such as pressure, shear force, and surface texture in real time. In parallel, a multichannel electrical stimulator and a flexible stimulation armband were designed to accurately convert tactile information into spatiotemporally encoded and distinguishable electrical stimulation signals based on a biomimetic mapping strategy, thereby enabling multidimensional and intuitive haptic feedback. In prosthetic manipulation tasks simulated with a robotic arm, the BHFS successfully assisted users in performing delicate grasping and texture discrimination under audiovisual deprivation, effectively mitigating common issues such as excessive force and slippage. This work not only offers a system-level solution for establishing tactile feedback in prosthetics but also provides a new technical pathway for the development of wearable human-machine interactive systems.

2 | Results and Discussion

2.1 | BHFS Design and Architecture

To enhance intuitive tactile perception for prosthesis users and improve prosthetic controllability and embodiment, we developed a wearable closed-loop haptic interface known as the BHFS. This system integrates high-sensitivity triboelectric tactile sensing with multichannel TES. It consists of three tightly coupled modules: a TP-Glove for multidimensional tactile sensing, a flexible multichannel stimulator embedded in an armband for cutaneous feedback, and a real-time closed-loop controller that maps sensed inputs to stimulation commands (Figure 1a). During operation, the prosthetic hand responds to the user's movement intention, while the TP-Glove continuously converts normal pressure, shear force, and texture information into corresponding electrical signals. The controller encodes these signals and drives spatially and temporally patterned TES to cutaneous receptors, eliciting percepts that are routed to central pathways, thereby closing the sensing-feedback loop in real time (Figure 1a(i)). In addition to facilitating tactile restoration for prosthetics, this closed-loop paradigm also supports teleoperation scenarios requiring intuitive force feedback between humans and haptic-deficient machines (Figure 1a(ii)).

Specifically, the detailed structure of the TP-Glove is depicted in Figure 1b. It incorporates a triboelectric-effect-based TFT-Sensor, with a triboelectric layer composed of $SrTiO_3$-doped Polydimethylsiloxane (PDMS), designed with a pyramid microstructure to significantly enhance output performance. The device's base is made from thermoplastic polyurethane (TPU), and the electrodes are fabricated using stretchable conductive silver paste, which serves as both an additional triboelectric layer and a conductive layer (Figures S1 and S2). When external pressure or tangential forces are applied, electrical signals are generated through triboelectrification and electrostatic induction. Meanwhile, utilizing a body-specific domain-matching strategy and phantom limb neuroprinciples, a spatiotemporal encoding scheme maps the captured sensing signals to corresponding electrical stimulation commands, achieving real-time closed-loop feedback through the "sensing–encoding–stimulation" process (Figure 1c). In addition, Figure 1d illustrates the structure of the

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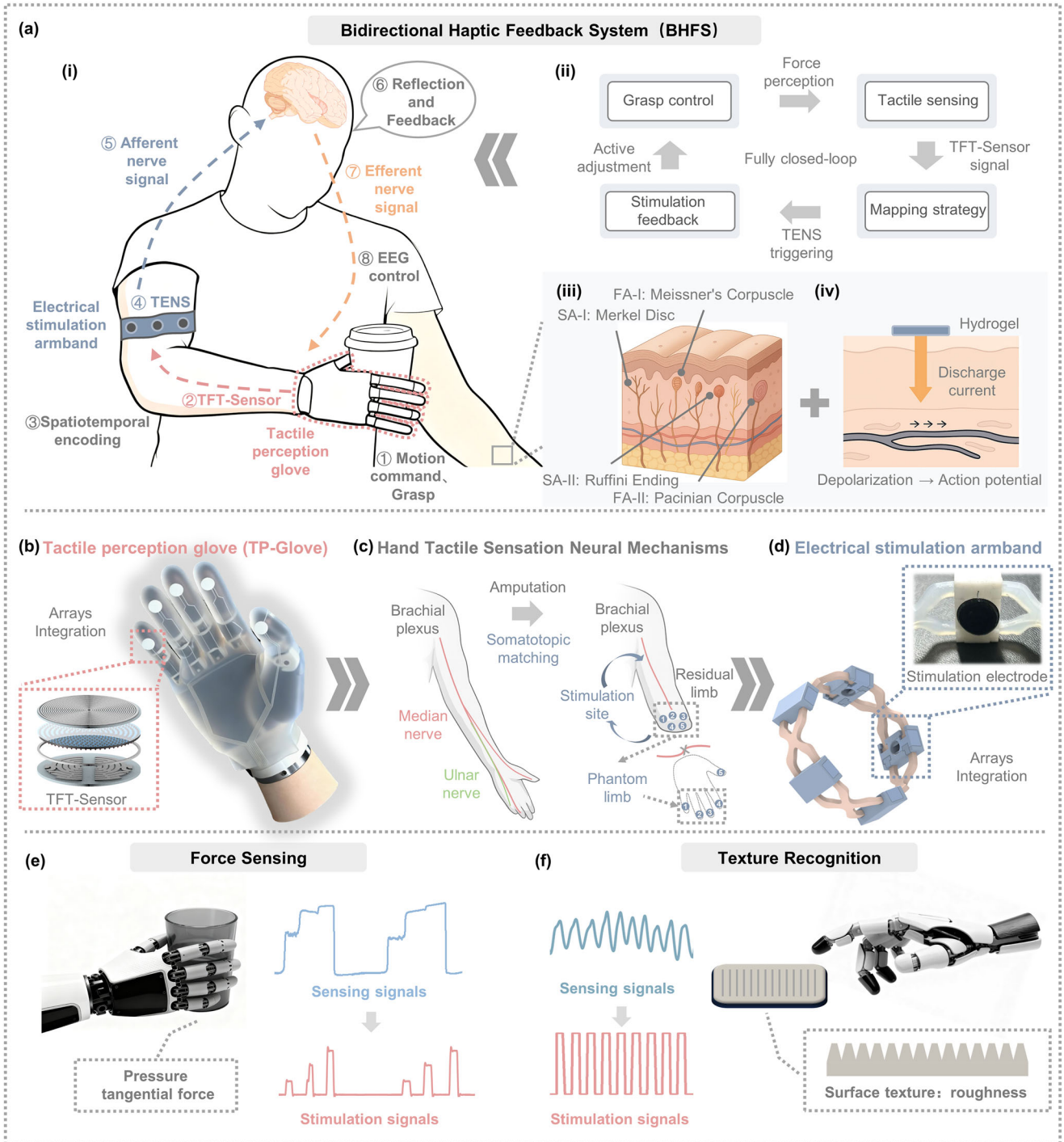


FIGURE 1 | System overview and design of the BHFS. (a) Schematic diagram of system components and closed-loop mechanism: (i) Sensing-feedback closed-loop pathway; (ii) Closed-loop feedback mapping process; (iii) Physiological basis of finger tactile perception; (iv) Neural mechanisms of tactile sensation induced by electrical stimulation. (b) Tactile perception: Flexible glove (TP-Glove) integrated with a Triboelectric Flexible Tactile Sensor array. (c) Perception-stimulation mapping strategy based on body-specific domain matching. (d) Stimulation feedback: Flexible armband driven by a multichannel electrical stimulator. (e) Real-time sensing and feedback of grip strength during grasping. (f) Recognition of surface textures during slipping.

stimulation feedback module, which includes a multichannel constant-current output system capable of delivering stable electrical stimulation feedback based on the mapped sensing signals. The flexible armband contains five independent stimulation electrode points, each corresponding to a specific sensor within the array. These circular electrodes are coated with hydrogel patches that broaden current distribution, reduce localized current peaks, and enhance both comfort and tactile sensitivity. Through the integration of these modules, the BHFS offers real-time perceptual feedback during routine tasks such as object grasping or texture recognition, significantly enhancing the controllability and interaction experience of prosthetic hands (Figure 1e,f).

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2.2 | Optimization and Electrical Output Characteristics of the TFT-Sensor

Utilizing the self-powering capability and high sensitivity of the triboelectric effect, we engineered the TFT-Sensor and systematically optimized its key performance parameters. The TFT-Sensor functions through triboelectrification and electrostatic induction: the contact and separation of two materials with differing electron affinities generate surface charge, which drives an external current via induced potential differences. The key parameters influencing output include the dielectric properties of the triboelectric layer, surface microstructures, and electrode geometry [53–56].

We fabricated and compared multiple sensor configurations using the experimental setup shown in Figure S3. Doping PDMS with $SrTiO_3$ raised the dielectric constant and substantially increased charge accumulation (Figure S4), while pyramid microstructures expanded the effective contact area; increasing electrode width improved charge collection efficiency. These combined optimizations elevated the sensor's open-circuit voltage (Voc) from about 8 to 14 V (Figure 2a; Figure S5). The optimized device, consisting of $SrTiO_3$-modified PDMS with pyramid patterns and 1 mm electrodes, was produced through molding, ionic bonding, and screen printing, resulting in a fully integrated, mechanically compliant sensing unit (Figure 2b). Scanning electron microscope (SEM) and surface imaging confirmed the precise replication of concentric fingerprint-like ridges and uniform pyramid arrays (Figure S6). Additionally, COMSOL simulations revealed the electric field distribution and working mechanism under a single-electrode mode, validating the rationale behind the structural design (Figure 2c; Figure S7).

The electrical performance of the TFT-Sensor was systematically characterized. The voltage output of the sensors increased with applied displacement, rising from 4 to 14 V (Figure 2d). Across various loading frequencies, both the Voc and transferred charge (Qsc) remained approximately constant (~14 V and ~4 nC), while the short-circuit current (Isc) increased with frequency from ~20 to ~70 nA (Figure 2e; Figure S8). The sensor demonstrated rapid response and recovery times (43.3 and 46.2 ms, respectively) and maintained clear signal fidelity up to 100 Hz (effective response ~10.1 ms), facilitating the detection of rapid mechanical events (Figure 2f; Figure S9). Mechanical robustness was evidenced by negligible degradation after repeated twisting/stretching and following 10 000 fatigue cycles under the conditions of 0.5 N and 1.5 Hz (Figure 2g; Figure S10). Environmental stability tests show that the sensor maintains a stable output voltage above 11 V within 20%–50% relative humidity (typical indoor conditions). Even at high humidity levels (>60% RH), the output waveform remains clear and distinguishable with a usable signal-to-noise ratio. Across 20°C–50°C, the output voltage increases only slightly with temperature, far smaller than the pressure-response range, indicating good temperature stability. Furthermore, no significant waveform distortion is observed in a dusty environment (Figures S11–S13).

The output amplitude increased with applied force and reached saturation at 50 g. The sensor achieved a pressure detection limit of 0.2 g (2 mN), sufficient for detecting the micro-forces involved in daily fine grasping tasks. Within the effective sensing range of 0.2–50 g, the sensor exhibited a sensitivity of 29.3 V/N, with output voltage positively correlated with the applied force (Figure 2h; Figures S14 and S15). Finally, a dual-channel electrode layout with PTFE isolation ridges effectively mitigated crosstalk and enabled independent tangential force detection (Figure 2i; Figure S16). Based on this, normal pressure can be calculated from the average of the two channel signals, and tangential shear force can be obtained from the differential signal, thereby achieving real time decoupling of the two forces (see Methods for detailed calculation).

In summary, the TFT-Sensor exhibits exceptional structural design, electrical performance, and mechanical robustness, establishing a strong foundation for multidimensional, high-sensitivity tactile sensing.

2.3 | TP-Glove Integration and Functional Evaluation

After validating the structural design and performance advantages of the individual sensing units, the array was integrated into a flexible glove to facilitate multidimensional tactile perception. Considering that the fingertips have a dense distribution of mechanoreceptors and high spatial resolution, and are the primary sites for fine tactile perception and precise dexterous grasping, we fabricated a stretchable glove equipped with screen-printed electrodes and strategically integrated five TFT-Sensor units at the fingertip locations to create the TP-Glove (Figure 3a; Figure S17). This arrangement strategy aims to validate the core functionality of the closed-loop system with controllable complexity. A custom-designed acquisition circuit (Figure S18) incorporating a charge-retained circuit and wireless module enabled stable, real-time transmission of triboelectric signals [57] (Figure S19). A testing platform comprising a robotic hand, acquisition system, and visualization interface was established to assess pressure, shear force, and surface texture recognition (Figure 3b).

The response to pressure was characterized initially. As illustrated in Figure 3c, the TP-Glove demonstrated sensitive, positively correlated voltage responses to repeated fingertip pressing. Under sustained loading, it effectively distinguished between the pressing, holding, and releasing stages (Figure 3d) while facilitating real-time visualization (Figure 3e; Video S1). These findings indicate reliable real-time monitoring of both static and dynamic pressure. In multi-finger grasping tests, the glove captured instantaneous multichannel responses (Figure S20) and tracked continuous force variations (Figure S21). Stable outputs from the thumb, index, and middle fingers confirmed the feasibility of multi-finger tactile sensing (Figure S22). Triboelectric sensors inherently generate pulsed signals. Therefore, a charge-retention circuit developed in our previous work was employed to integrate the generated charge, enabling stable, continuous output under sustained or slowly varying tactile inputs [57].

The dual-channel structure of the TFT-Sensor further enabled shear force monitoring. As depicted in Figure 3f,g, the two

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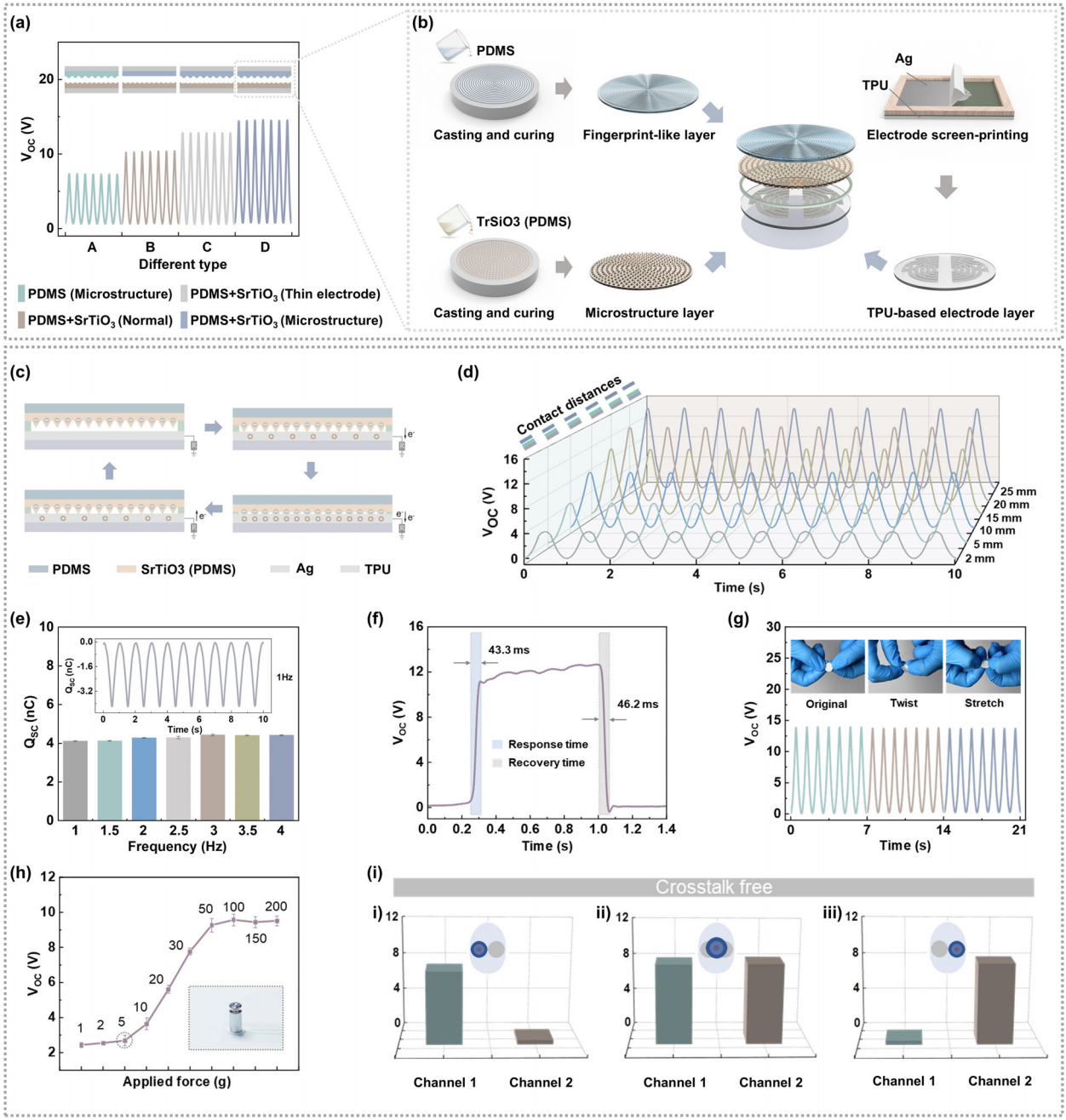


FIGURE 2 | Optimization and electrical output performance of the TFT-Sensor. (a) Influence of material composition and surface microstructures on output voltage. (b) Fabrication process of the optimized TFT-Sensor. (c) Working principle under single-electrode mode. (d) Output voltage as a function of applied displacement. (e) Effect of loading frequency on charge transfer. (f) Response and recovery times. (g) Stability test under mechanical deformation. (h) Relationship between output signals and applied force. (i) Dual-channel electrode design with PTFE isolation for independent force detection and crosstalk suppression.

channels exhibited opposite responses under upward and downward dragging, with their differential signal representing the magnitude and direction of shear force. Signals from the index finger sensor corroborated this relationship (Figures S23 and S24). During dual-finger tests, the TP-Glove accurately tracked variations in shear force magnitude and direction in real time (Figure 3h; Video S2).

For surface texture recognition, the robotic hand was driven to slide across substrates with varying roughness levels. The frequency of the triboelectric signal decreased with increasing roughness, while finer textures produced higher-frequency outputs (Figure 3i). We constructed a convolutional neural network (CNN) classifier, comprising 4 convolutional layers, 3 pooling layers, and 2 fully connected layers (see Methods for detailed data acquisition, dataset configuration, and CNN training parameters), which takes time-series signal segments from the triboelectric sensor as input and achieves an accuracy exceeding 97% in distinguishing textures (Figure 3j; Figure S25). Real-time sliding tests (Video S3) confirmed precise online

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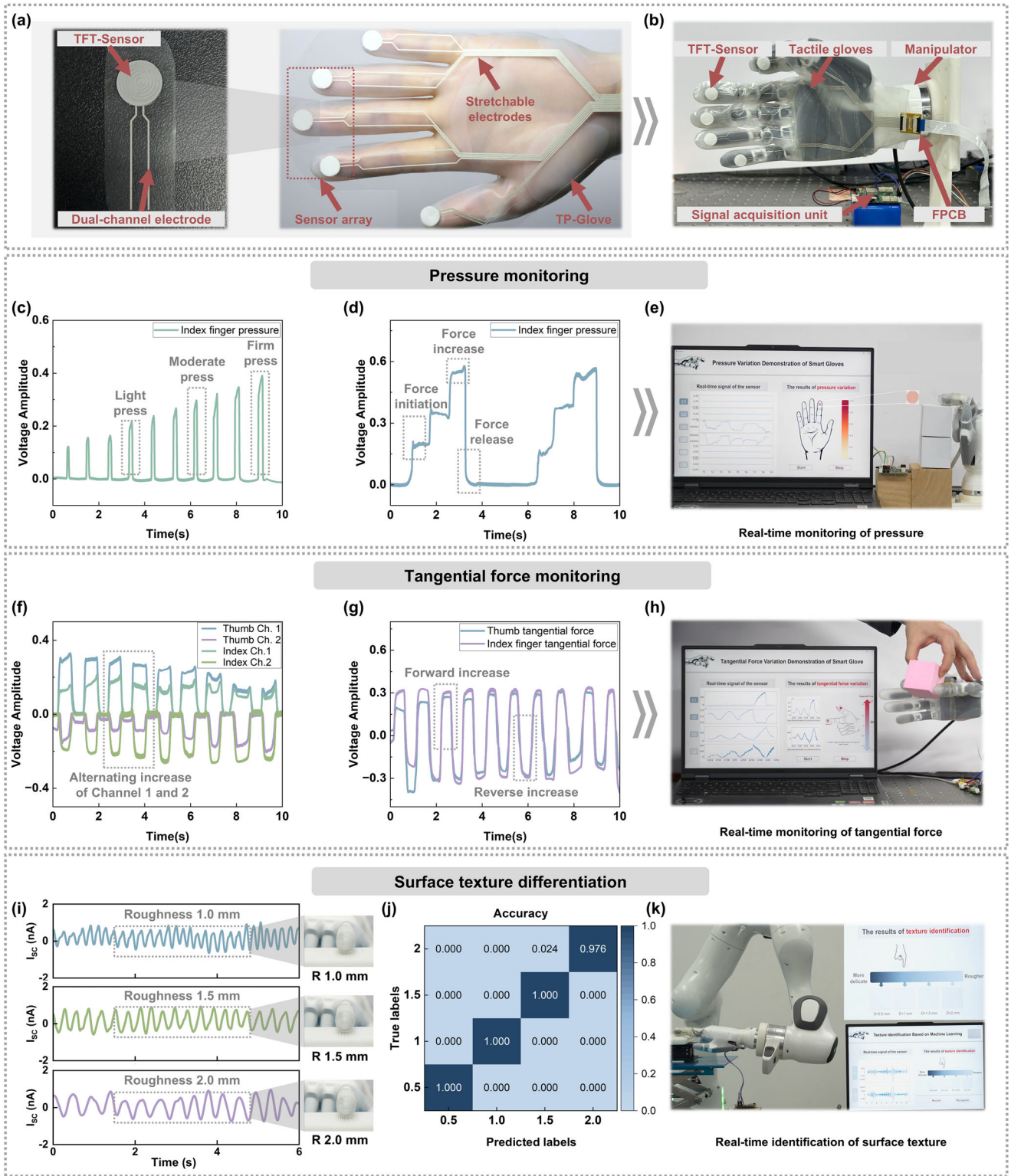


FIGURE 3 | Integration and functional evaluation of the TP-Glove. (a) Integration of the TFT-Sensor array into the flexible glove. (b) Testing platform for tactile sensing. (c–e) Single-finger pressure sensing: transient response, continuous monitoring, and visualization. (f, g) Raw signals and differential outputs under dual-finger shear forces. (h) Real-time detection of shear forces. (i) Triboelectric signals generated from sliding over surfaces with varying roughness. (j) CNN-based classification accuracy for surface roughness. (k) Screenshot from the real-time interface illustrating successful online texture recognition during sliding.

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recognition, with Figure 3k displaying representative results. Notably, the TFT-Sensor can also resolve fine textures down to 0.3 mm (Figures S26 and S27), demonstrating its capability for submillimeter-scale texture discrimination, laying the foundation for achieving finer scale detection and thus better matching the sensing capability of human skin.

In summary, the TP-Glove demonstrated sensitive detection of normal and shear forces with stable continuous monitoring, and accurately distinguished surface textures with high resolution. These results underscore its significant potential for tactile feedback in prosthetic applications and human-machine interaction.

2.4 | Multichannel Electrical Stimulator

TES provides a non-invasive, versatile, and rapidly responsive approach to eliciting cutaneous sensations, making it a compelling option for delivering naturalistic and personalized haptic feedback. To facilitate coordinated feedback between the sensing glove and the prosthesis, a multichannel electrical stimulator has been developed (Figure S28). Its core components include a microcontroller, a constant-current boosting unit, and a channel management unit (Figure S29). The constant-current design accommodates variations in skin impedance to ensure stable output, while the channel management unit supports up to six independently controlled channels. Redundancy in circuit design offers flexibility for multidimensional feedback experiments. Considering the effects of waveforms on stimulation efficacy and comfort, pulsed square waves were selected as the primary output. Under simulated skin impedance of 10 kΩ at a constant frequency, the stimulator delivered stable currents ranging from 0.5 to 10 mA, satisfying graded feedback requirements while minimizing skin irritation (Figure 4a). Frequency adjustments within the 1–5 Hz range reproduced diverse tactile sensations and reduced adaptation (Figure 4b). Duty cycle modulation (10%–50%) further refined stimulus intensity beyond amplitude control, enabling subject-specific tuning for optimal comfort (Figure 4c). In addition, a 2 h continuous discharge test showed no significant degradation in output performance, confirming the long-term output stability of the stimulator (Figure S30). Collectively, these adjustable parameters provide a robust foundation for multimodal feedback.

To assess perceptual responses, six subjects were recruited under audiovisual shielding and presented with randomly applied stimulation patterns across 20 repeated trials (2 s each), with instructions to rate their perceptions. In amplitude tests, subjects consistently distinguished five levels ranging from “no sensation” to “pain,” although thresholds varied across participants (Figure 4d). Frequency recognition achieved >90% accuracy across “slow-medium-fast” levels (Figure 4e). Preferences for pulse width varied among subjects; notably, the 50% duty cycle was rated as the most comfortable by over 80% of participants (Figure 4f). These results confirm the reliability of graded stimulation while highlighting the necessity for individualized calibration to enhance perceptual consistency. Detailed experimental setups are outlined in the Methods section.

For effective delivery, a flexible, noninvasive stimulation armband was designed to conform to the arm and integrate seamlessly with the stimulator (Figure 4g, Figure S31). Beyond parameter tuning, spatiotemporal encoding of stimulation points enriched feedback strategies. To evaluate spatial perception, stimulation was randomly delivered to one to three of the five electrode channels, defining the activation patterns for encoding (Figure 4h). Subjects discriminated between single-channel, dual-channel, and multi-channel stimulation with an average accuracy of 84.7% (Figure 4i). To assess temporal perception, distinct activation sequences (e.g., sequential, reverse, jumping) were applied across three electrodes, implementing predefined temporal encoding schemes (Figure 4j). Subjects distinguished random sequence patterns (sequential, reverse, jumping, and returning) with an average accuracy of 84.8% (Figure 4k). These findings indicate that a compact combination of spatial and temporal codes can effectively convey multidimensional tactile information via noninvasive TES. Specific experimental protocols are detailed in the Methods section.

2.5 | BHFS Construction and Prosthetic Tactile Feedback Applications

Building upon the TP-Glove and flexible stimulation armband, we developed the BHFS to address the deficiency of intuitive tactile feedback in prosthetic hands, which hinders amputees’ ability to perceive and control hand movements. The system consists of three integrated modules: tactile sensing (TP-Glove and acquisition unit), closed-loop control (real-time signal mapping), and feedback stimulation (electrical stimulator and flexible armband), forming a complete “sensing-feedback” loop (Figure 5a; Figure S32).

The core of the closed-loop control is a spatiotemporal mapping strategy that converts multidimensional tactile cues into patterned electrical stimulation. For force sensing, normalized signal amplitudes were discretized into graded levels and mapped to the amplitude of electrical stimuli in the temporal domain, while each sensor channel was assigned to a dedicated electrode to provide somatotopic feedback. For texture recognition, triboelectric signal frequencies were discriminated through filtering and mapped to distinct stimulation frequencies within the same spatial scheme. This integrated approach enabled the system to effectively interpret and convey a rich set of tactile information, including static (pressure), dynamic (texture), and their spatiotemporal variations as intuitive spatiotemporal stimuli.

To evaluate the system’s efficacy, grasping experiments were conducted with individualized stimulation parameters (Figure S33). The robotic hand equipped with the TP-Glove provided real-time pressure data, while subjects received corresponding feedback via the armband. Control software filtered signals and triggered stimulation upon reaching preset thresholds. During repeated grasps, the system accurately captured pressure variations and generated synchronous, instantaneous stimulation responses (Figure 5b). In sustained grasps, pressure signals increased stepwise and remained stable, while stimulation intensity scaled accordingly, demonstrating continuous and controllable feedback (Figure 5c).

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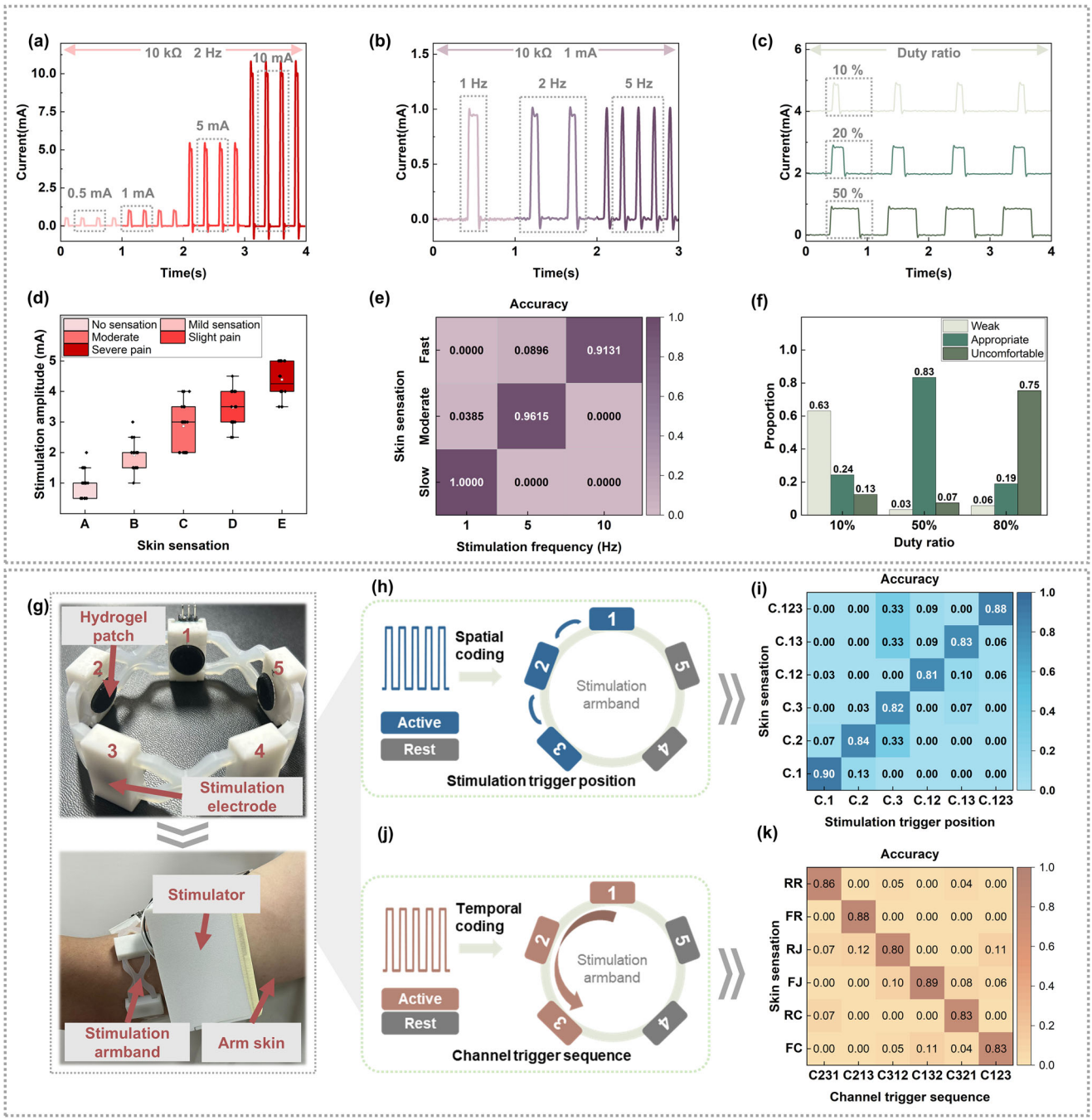


FIGURE 4 | Design, characterization, and validation of the electrical stimulator. (a) Output currents measured under a 10 kΩ impedance at a frequency of 2 Hz. (b) Output characteristics across frequency ranges of 1–5 Hz. (c) Duty cycle modulation (10%–50%) assessed under a 10 kΩ impedance. (d) Subjective ratings of amplitude perception. (e) Results of frequency discrimination. (f) Comfort ratings associated with various duty cycles. (g) Photograph of the flexible armband. (h) Spatial encoding through random electrode activation. (i) Accuracy in spatial discrimination. (j) Temporal encoding represented by activation sequence schemes. (k) Accuracy in temporal discrimination(Forward Complete, Reverse Complete, Forward Jump, Reverse Jump, Forward Return, Reverse Return).

Closed-loop functionality was further validated using a robotic arm simulating prosthetic grasping. Subjects, blindfolded and wearing noise-canceling headphones, performed tasks relying solely on tactile feedback (Figure S34). In the control group, the absence of feedback impeded precise force control, often resulting in object damage (Figure 5d,e). In the feedback group, stimulation amplitude scaled with initial force, prompting subjects to adjust their grip; as pressure decreased, stimulation weakened, completing the feedback loop and enabling stable object grasp without damage (Figure 5f,g; Video S4). These results demonstrate that the BHFS effectively supports precise grasp control under sensory-isolated conditions.

To assess tactile interaction, a texture discrimination task was performed. The robotic hand, equipped with the TP-Glove, slid

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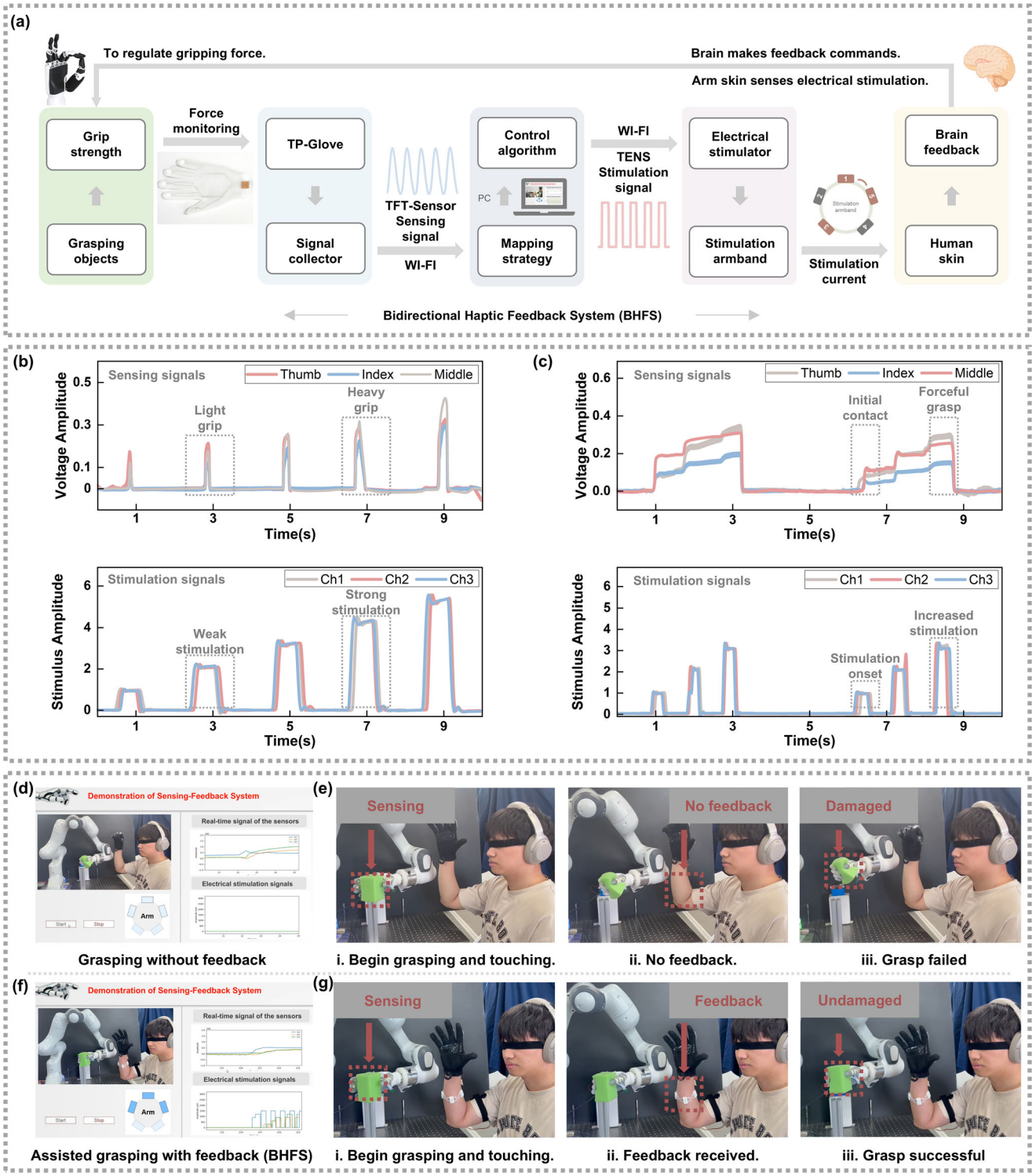


FIGURE 5 | Construction of the BHFS and Its Application in Prosthetic Tactile Feedback. (a) Schematic representation of the BHFS architecture and closed-loop control system for prosthetic tactile feedback. (b) Response of the "sensing-feedback" mechanism during transient grasping. (c) Response of the "sensing-feedback" mechanism during sustained grasping. (d) Control group: grasping without feedback. (e) Control group: failure to grasp the object. (f) Feedback group: BHFS-assisted grasping. (g) Feedback group: stable object grasp.

across surfaces of varying roughness, with triboelectric signals mapped to stimulation frequency (Figure S35). High-density textures elicited 5 Hz stimulation, prompting subjects to extend the hand, whereas low-density textures generated 1 Hz stimulation, eliciting a grasping gesture (Figure S36; Video S5). This confirms that users could reliably perceive surface roughness via BHFS feedback.

In summary, the BHFS establishes a robust closed-loop system for multidimensional tactile sensing, real-time signal mapping, and

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intuitive stimulation feedback. Even in the absence of visual or auditory cues, subjects achieved precise grasp control and texture discrimination, underscoring the system's potential for prosthetic tactile restoration and bidirectional human-machine interaction.

3 | Discussion

This study introduces a flexible BHFS that integrates triboelectric sensing and TESto address the core bottleneck in current prosthetic hands: the absence of effective sensing and feedback mechanisms leading to non-intuitive operations and unnatural interactions. Unlike previous research that typically focuses on optimizing individual sensing or feedback components, this work develops a fully integrated, wearable closed-loop system. By fusing multidimensional tactile perception with tunable electrical stimulation feedback, it systematically addresses long-standing challenges, including limited sensing dimensionality, inadequate feedback naturalism, and poor system integration.

In terms of system design, the BHFS enables high-accuracy perception of pressure, shear force, and surface texture through a flexible sensing glove, achieving a texture recognition rate of 97%. When integrated with multi-parameter tunable TES, it significantly outperforms conventional vibration or pneumatic approaches in both feedback information dimensionality and naturalism. Crucially, the "sensing-stimulation" mapping strategy proposed in this study converts tactile information into spatiotemporally encoded electrical stimulation feedback. This enables users to accurately discern grip force, detect impending slip, and identify surface textures even without audiovisual assistance—an integrated perceptual capability that has not been adequately realized in existing prosthetic systems.

From a technological perspective, the TFT-Sensor is particularly well-suited for detecting minute force variations in prosthetic manipulation due to its high sensitivity, self-powering capability, and superior signal-to-noise ratio. Meanwhile, TES technology, characterized by its non-invasive nature and flexible, tunable parameters, offers significant advantages in user safety and experience compared to invasive methods. More importantly, by dynamically modulating stimulation parameters and spatiotemporal encoding, the system demonstrates potential for high-fidelity reconstruction of lost tactile neural pathways in amputees. This advancement not only enhances operational precision but also has profound implications for restoring the sense of embodiment, alleviating phantom limb pain, and promoting social integration, thereby delivering significant physiological and psychological rehabilitation benefits.

While this study has successfully demonstrated robust system construction and experimental validation, several avenues warrant further exploration. Future research should include long-term clinical validation with amputee populations, with systematic evaluation based on the principle of phantom limb mapping. At the same time, bio-inspired electrical stimulation algorithms that accommodate individual perceptual differences should be developed to further enhance the naturalness of feedback. Expanding the perceptual coverage to the palm and interphalangeal joint areas can further improve the completeness of tactile restoration. These research directions are key to enhancing the sensing and feedback capabilities of the system and will help construct a biomimetic tactile feedback system that more closely mimics the natural perceptual abilities of the human hand. Furthermore, advancing deep integration with commercial myoelectric prostheses and optimizing system power consumption and form factor are critical steps toward translating this technology into practical applications.

In summary, this research not only provides a systematic solution for tactile functional restoration in prosthetics but also establishes a technical foundation for the next generation of wearable human-machine systems through its core principle of achieving natural interaction via multidimensional perception-feedback mapping. The design philosophy and implementation methodology of this system offer valuable insights for fields requiring high-fidelity haptic feedback, such as teleoperation and virtual/augmented reality. This work reflects both the social value of technology serving human needs and demonstrates significant economic potential and application prospects in the consumer electronics and digital health industries.

4 | Methods

4.1 | Fabrication of the TFT-Sensor

PDMS prepolymer and curing agent (Sylgard 184, Dow Inc., 10:1) were mixed with hexane (Macklin Inc.) and $SrTiO_3$ powder (Sigma–Aldrich Inc., 10:0.66). The mixtures were stirred (600 rpm, 15 min), degassed (15 min), and spin-coated onto lithographic molds (45 rpm, 15 s). After curing (60°C, 5 h), pyramid and fingerprint-structured films were demolded, plasma-treated (45 s), and bonded under pressure at 60°C (3 h) to form the top triboelectric layer.

Stretchable silver electrodes were patterned by screen-printing silver paste (DuPont Inc.) onto TPU films (McMaster-Carr Inc.), followed by curing (60°C, 4 h) to yield the bottom electrode layer. A silicone ring (UV-curable adhesive, Xining Tec Inc.) was attached, and the top triboelectric layer was bonded (Sil-Poxy, Smooth-On Inc.). Finally, Ecoflex30 (Smooth-On Inc.) was spin-coated as an encapsulation layer, degassed, and cured (60°C, 2 h). The fully packaged device functioned as a single-electrode triboelectric sensor.

4.2 | Characterization of the TFT-Sensor

SEM images were taken by a field emission SEM (SU 8020, Hitachi). Optical photographs were taken by a digital camera (A7M4, Sony). The TFT-Sensor was driven using a linear motor (E1100, LinMot) for basic data acquisition (frequency response test, mechanical response test, fatigue test, etc.). The output performance of the TFT-Sensor (open-circuit voltage, short-circuit current, and transferred charge) was tested using an electrometer (Keithley 6514, Keithley Corp.) and recorded by LabView with a data acquisition card (USB-6009, NI Corporation). Mechanical calibration was performed using a force transducer (AD2016SC2, Autoda).

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4.3 | Data Acquisition System

The voltage signals from the triboelectric sensor were first amplified by precision operational amplifiers. The conditioned analog signals were digitized using dual ADS131M04 analog-to-digital converters (ADCs) and transmitted to an ESP32 microcontroller. The ESP32 packaged the data into serial strings and wirelessly transmitted them to a computer via Wi-Fi. A Python-based multithreaded TCP server program processed the incoming data in real time, enabling synchronized acquisition, visualization, and storage with high accuracy and reliability.

4.4 | Data Collection and Dataset Configuration for the Texture Recognition Experiment

A robotic arm (Franka Emika Panda) was used to drive the TP-Glove to slide over textured plates at a constant speed and contact pressure. The TFT-Sensor on the index finger of the TP-Glove generated sensing signals, which were acquired by the signal acquisition circuit board. Four textured plates with surface roughness levels of 0.5, 1.0, 1.5, and 2.0 mm were used. To simulate diverse contact conditions, eight sliding postures were designed. The sampling frequency was set to 50 Hz, and each acquisition lasted 2 s. For each combination of texture and posture, 100 repeated acquisitions were performed, yielding 100 samples per combination. The total sample size was 4 textures $\times$ 8 postures $\times$ 100 repeats = 3200 samples. The acquired signals were normalized to the [0, 1] range and then randomly split into a training set (70%) and a test set (30%).

4.5 | Decoupling Calculation Method for Pressure and Shear Force

Let the output voltages of Channel 1 and Channel 2 be V_1 and V_2, respectively. The normal pressure P is estimated from the mean of the two channel signals through an experimentally calibrated pressure–signal relationship: $P = f_p\left(\frac{V_1+V_2}{2}\right)$. The magnitude of the tangential shear force F_s is estimated from the absolute value of the differential signal through an experimentally calibrated shear–signal relationship, and the direction is determined by the sign of the difference: $F_s = f_s(|V_1 - V_2|)$, $direction = sign(V_1 - V_2)$, The calibration functions f_p and f_s were obtained through independent calibration experiments.

4.6 | Fabrication of the Electrical Stimulation Armband

The armband comprised stimulation electrodes, a 3D-printed case (P1S, Bambu Lab Inc.), hydrogel patches, and a silicone connector. Dragon Skin 30 silicone (Smooth-On Inc., parts A and B mixed 1:1) was mixed (500 rpm, 1 min), degassed (5 min), cast into 3D-printed molds with pre-embedded electrode wires, and cured (40°C, 4 h) to form the connector ring. Metallic disk electrodes were soldered to the wires for hydrogel attachment, and five electrodes were connected to a PCB for current delivery. Commercial medical-grade hydrogel patches were applied to the metal electrodes and can typically be used for approximately 3–5 days under normal use conditions (2–3 h per day) before replacement. In addition, the hydrogel patches are low-cost consumables, with each pair costing approximately USD 0.3-0.4.

4.7 | Characterization of Electrical Stimulation

Stimulator output currents were measured using an electrometer (Keithley 6514), and signal waveforms were recorded with an oscilloscope (MSO 2024B, Tektronix).

4.8 | Psychophysical Characterization of Electrical Stimulation Perception

All psychophysical experiments were conducted with six participants under audiovisual shielding. For each parameter set, 20 randomized trials were delivered per participant, with each stimulus lasting 2 s. Perception accuracy was calculated by comparing participant reports against the actual stimulation parameters. The specific paradigms were as follows:

4.8.1 | Amplitude

Five current levels (1.0, 2.0, 3.0, 4.0, 5.0 mA) were mapped to subjective intensity ratings ranging from “No sensation” to “Severe pain.”

4.8.2 | Frequency and Duty Cycle

Three frequencies (1, 5, 10 Hz) and three duty cycles (20%, 50%, 80%) were evaluated, corresponding to perceived levels of “Slow-Moderate-Fast” and “Weak-Appropriate-Uncomfortable,” respectively.

4.8.3 | Spatial Perception

Stimulation was randomly delivered to one, two, or three of the five electrode channels. Participants identified the activated site(s).

4.8.4 | Temporal Perception

Stimulation was applied to three electrodes in distinct spatiotemporal sequences (e.g., 1-2-3, 3-2-1, 1-3-2). Participants reported the perceived sequence pattern.

4.9 | Application Experiments

Multidimensional force sensing of the TP-Glove: The TP-Glove was mounted on a robotic hand (RH56BFX-2, INSPIRE-ROBOTS) to evaluate: (i) normal force detection by varying the bending angles against a metal block, (ii) shear force detection by dragging the block vertically during grasping, and (iii) slip detection using a robotic arm (Panda, Franka Emika) to perform reciprocating sliding on textured plates.

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The effect of electrical stimulation feedback: Participants established reference sensations for (i) amplitude (1.0–5.0, 0.5 mA increments, five levels), (ii) frequency (1–10, 5 Hz increments, three levels), and (iii) duty cycle (20%–80%, three levels). For each parameter, 20 randomized stimuli were delivered, and subjective ratings were recorded.

Closed-loop strategies for TP-Glove-based multichannel electrotactile feedback: In grasping tasks, participants controlled the robotic hand using the TP-Glove. Without feedback, the robotic hand simply mirrored motions; with the electrical stimulation armband, stimulation triggered upon object contact enabled participants to modulate their grasp force. In texture discrimination tasks, sliding the TP-Glove on low and high roughness plates was mapped to 1 and 5 Hz stimulation, respectively, allowing participants to distinguish textures through feedback. All trials were performed with participants blindfolded and wearing headphones. Informed consent was obtained from all human participants prior to their inclusion in the study.

Author Contributions

J.C., X.Q., and Y.S. supervised the project and were responsible for conceptualization. X.Q., H.Z., and B.Y. developed the methodology. B.Y. and H.Z. implemented the software. X.Q., H.Z., and X.K. designed and set up the hardware. H.Z., B.Y., X.C., T.C, and M.L. conducted the investigation and data collection. X.Q., H.Z., B.Y. and X.H., performed data visualization. L.J., B.T., Z.L., J.S., Z.Y., and Y.J. assisted with the interpretation and description of the results. J.C. and X.Q. acquired funding. J.C. and Y.S. managed project administration. The original draft was written by H.Z., X.Q., and B.Y.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adfm76572-sup-0001-SuppMat.docx.

Supporting File 2: adfm76572-sup-0002-VideoS1.mp4.

Supporting File 3: adfm76572-sup-0003-VideoS2.mp4.

Supporting File 4: adfm76572-sup-0004-VideoS3.mp4.

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